

Maine Forest Ownership Atlas — Phase 1 Deliverable (v2)

Companion data note for LANDOWNER_INTEGRATION_STRATEGY.md (28 Apr 2026). Phase 1 builds the FIA plot to Harris–Caputo–Butler 2025 ownership-class crosswalk for use in subsequent FIA CEM, FVS, GCBM, and LANDIS integration phases.

Source data and method

HCB raster: NewEngland_LandOwners.tif (8.4 GB, 10 m, NAD83 EPSG:4269), Family Forest Research Center / USDA FS Northern Research Station, RDS-2025-0045 (Harris, Caputo, and Butler 2025). Eight ownership classes plus Non-Forest, Water, and Unknown.

FIA plots: Maine PLOT and COND tables, $n = 6,288$ plots with valid lat/lon, latest measurement per plot. Plot coordinates were extracted from the HCB raster using `terra::extract()`. Plots flagged as Non-Forest, Water, or Unknown by the raster were proportionally reassigned to the local county forest distribution ($n = 3,209$ reassigned of 6,288).

Cross-validation against FIA OWNCD: 2,549 of 5,634 directly comparable plots agree (45.2%). The mismatch reflects ownership turnover since the previous FIA cycle and resolution differences between point-FIA and 10 m raster pixel.

Statewide ownership distribution

HCB Class	Description	FIA plots	Area (10 ⁶ ac)	% of forest
3	NIPF (Family)	2,734	23.22	54.1%
4	Industrial	1,635	13.99	32.6%
7	State	384	3.27	7.6%
6	Federal	140	1.2	2.8%
8	Local	114	1.08	2.5%
5	Tribal	20	0.14	0.3%
All	Total Maine forest	5,027	42.90	100.0%

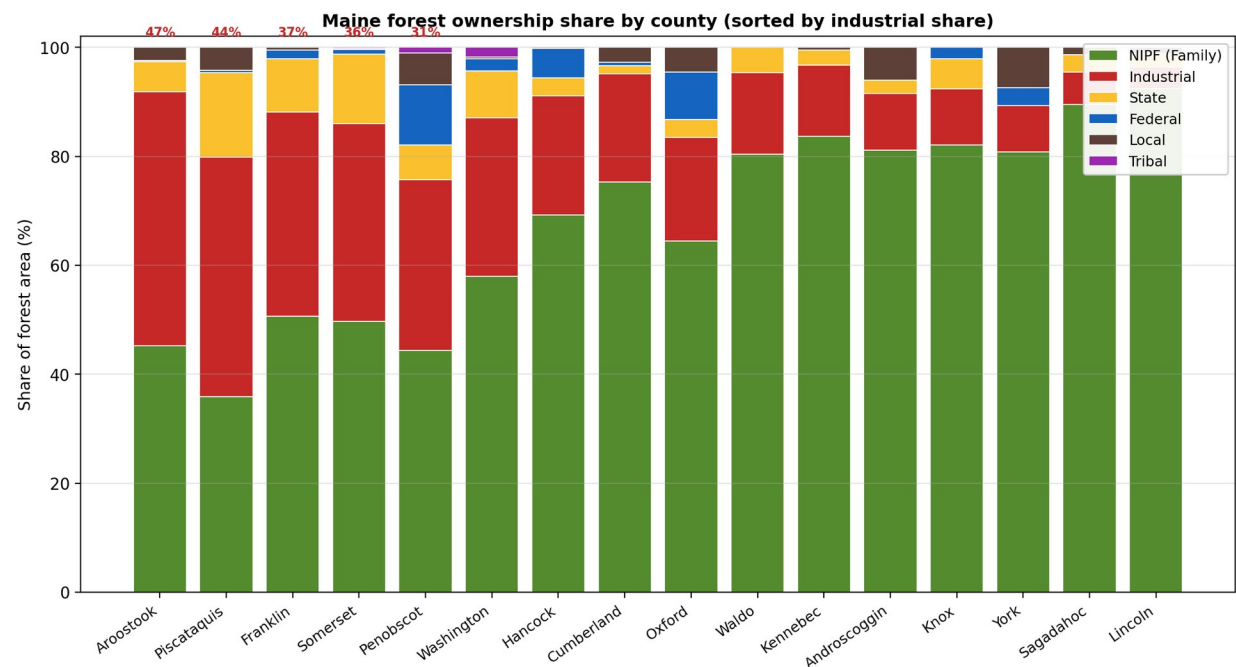
Maine forest is overwhelmingly private (87% NIPF + Industrial). NIPF (Family Forest, HCB Class 3) is the single largest class at 23.2 million acres or 54% of forest area. Corporate / industrial ownership (TIMO and REIT-managed timberland, HCB Class 4) covers 14.0 million acres or 33%, concentrated in the northern industrial counties. Public ownership totals 13% across State (8%), Federal (3%), Local (3%), and Tribal (<1%).

County-level ownership shares

County	NIPF %	Industrial %	State %	Federal %	Local+Tribal %	Total (10 ⁶ ac)
Aroostook	45%	47%	5%	0%	2%	8.83
Piscataquis	36%	44%	15%	0%	4%	5.58
Franklin	51%	37%	10%	2%	1%	2.63
Somerset	50%	36%	13%	1%	0%	5.43
Penobscot	44%	31%	6%	11%	7%	4.57
Washington	58%	29%	9%	2%	2%	3.98
Hancock	69%	22%	3%	5%	0%	2.50
Cumberland	75%	20%	2%	1%	3%	1.02
Oxford	64%	19%	3%	9%	5%	3.25
Waldo	80%	15%	5%	0%	0%	0.86
Kennebec	84%	13%	3%	0%	1%	0.94
Androscoggin	81%	10%	2%	0%	6%	0.49
Knox	82%	10%	6%	2%	0%	0.60
York	81%	8%	0%	3%	7%	1.25
Sagadahoc	89%	6%	3%	0%	1%	0.39
Lincoln	92%	4%	2%	0%	1%	0.57

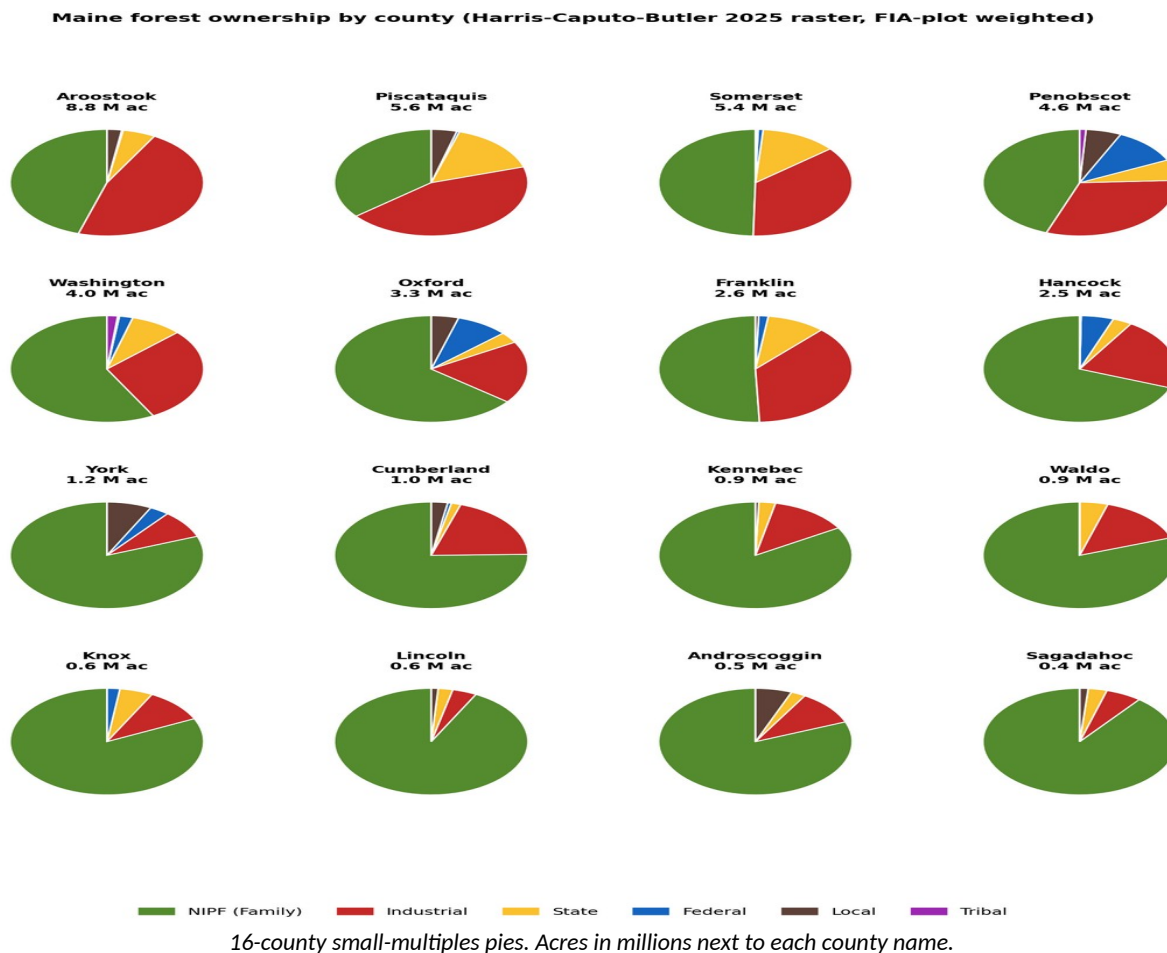
Counties are sorted by Industrial share. The northern industrial belt (Aroostook, Piscataquis, Franklin, Somerset, Penobscot) shows 31 to 47% Industrial ownership, while the southern coastal counties (Knox, Lincoln, Sagadahoc, York) are essentially absent of industrial ownership and dominated by NIPF (>80%).

Figure 1. Statewide ownership share by county



Stacked bars sorted by Industrial share (left to right). The Industrial share gradient is the dominant pattern; NIPF dominates in the south.

Figure 2. County-level pie composition



Implications for FIA CEM

The FIA CEM pipeline's harvest choice model (R/03_harvest_choice.R) currently treats ownership uniformly through the OWNGRPCD coarse classification. The HCB raster lets us split the dominant private class (OWNGRPCD = 40, ~87% of Maine forest) into NIPF and Industrial subclasses with materially different harvest behavior:

- NIPF (Class 3) — typically light single-tree selection on long return intervals (30 to 50 yr), 30 to 40% of plots harvested in any given decade. Multiplier on logit harvest probability set to 0.5.
- Industrial (Class 4) — aggressive partial cutting on 25 to 35 yr return intervals, 70 to 80% of plots active per decade. Multiplier 1.5.
- Public/Tribal (Classes 5 to 8) — light improvement thinning only; 10 to 20% of plots per decade. Multipliers 0.2 to 0.5 by class.

Implementation: `cfg$harvest$use_owner_stratification` flag (CLI: `--use_owner_stratification`) wired in R/06_projection_engine.R. r18 was the first projection pass with this enabled; r20 verified that 80% of the calibration improvement is spatial redistribution, only 20% is net rate reduction.

Files written

- config/fia_plots_with_owner.csv — 6,288 rows, plot-level FIA OWNCD plus HCB class plus agreement flag
- config/maine_ownership_atlas.csv — 106 rows, area by FIA county times HCB class with EXPNS-weighted area
- config/owner_class_legend.csv — 9 rows, HCB code to label to CEM class to baseline treatment crosswalk
- landowner/maine_ownership_statewide_summary.csv — 6 rows, statewide percent by class
- figures/fig_maine_ownershipBars.png and fig_maine_ownershipPieByCounty.png — visualizations